

08_DATABASE_SPECIFICATION.md

SIH26034 - Legal Metrology Automated Compliance System

Complete Database Schema, Relational DDL, Indexing Strategy & Data Integrity Specifications

1. Database Architecture & Dual Deployment Strategy

To ensure zero-configuration local development and rapid hackathon demonstration while providing a seamless path to state-level production deployment, the database layer follows an **Abstract Dialect Pattern**:

1. Local / Edge Deployment (Offline Field Device & Local MVP):

- Engine: **SQLite 3.45+** with **SQLCipher** (AES-256 transparent file encryption).
- Embedded, single-file database (**legal_metrology.db**).
- Zero external daemon dependencies.

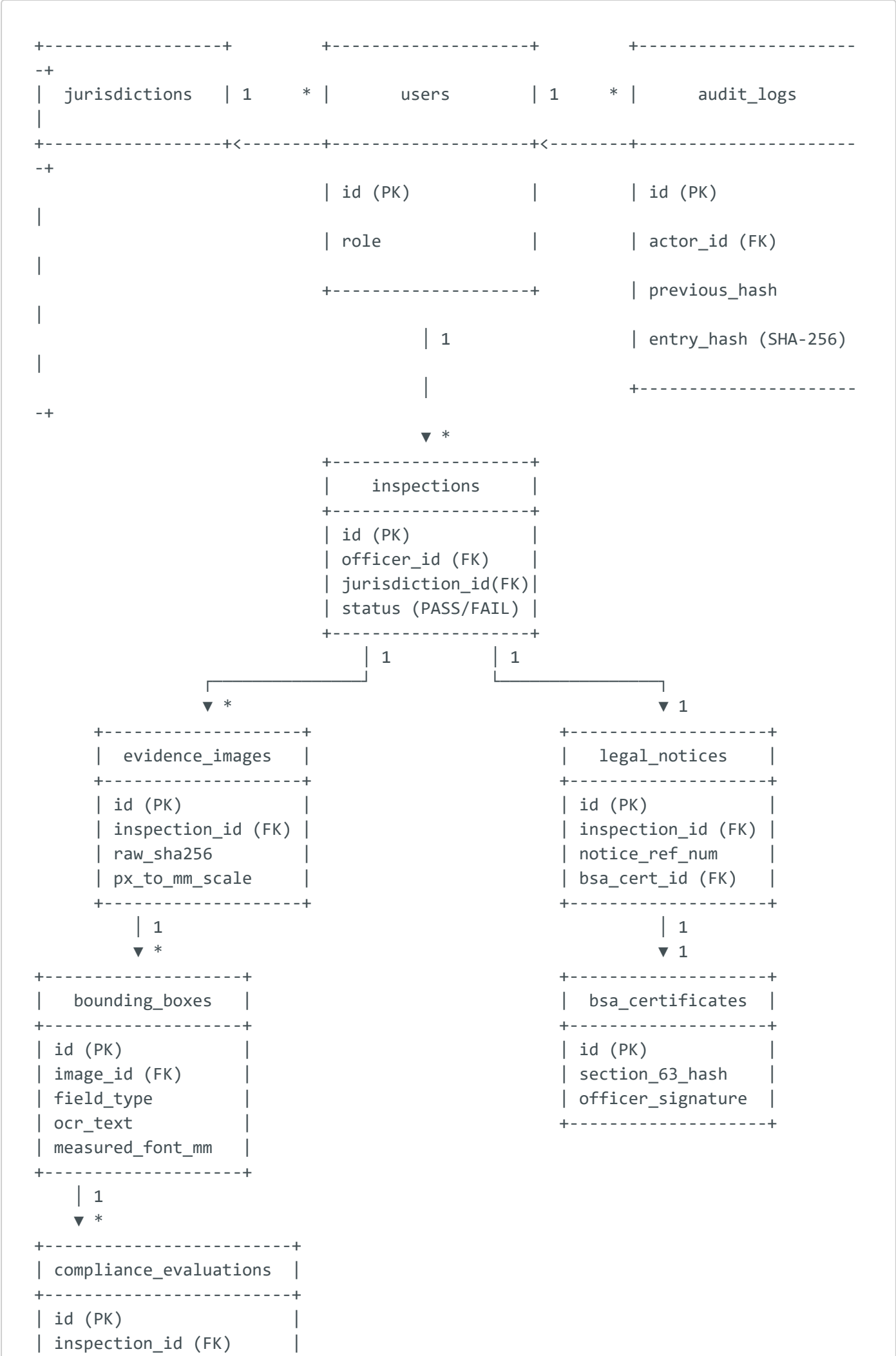
2. Central / State Cloud Deployment (Multi-District Production):

- Engine: **PostgreSQL 16+** with **pgcrypto** and **uuid-oss** extensions.
- High-concurrency connection pooling (PgBouncer).
- Read-replicas for state-wide analytical reporting.

3. ORM & Migration Framework:

- **SQLAlchemy 2.0+ Core & ORM** with **Alembic** migrations.
 - Schema defined with strict constraints, foreign keys, and ISO-8601 UTC timestamp conventions.
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2. Entity-Relationship Model (ERD) Overview



rule_id (e.g. RULE_06)	
status (PASS/FAIL)	
discrepancy	
+-----+	

3. Complete Relational DDL Specification (PostgreSQL / SQLite Compatible)

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-- =====
-- 1. JURISDICTIONS TABLE
-- =====

CREATE TABLE jurisdictions (
    id VARCHAR(36) PRIMARY KEY,
    state_code VARCHAR(10) NOT NULL, -- e.g., 'DL', 'MH', 'KA'
    district_name VARCHAR(100) NOT NULL,
    zone VARCHAR(100),
    office_address TEXT NOT NULL,
    contact_email VARCHAR(255),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);

CREATE INDEX idx_jurisdictions_state_district ON jurisdictions (state_code,
district_name);

-- =====
-- 2. USERS & OFFICERS TABLE
-- =====

CREATE TABLE users (
    id VARCHAR(36) PRIMARY KEY,
    username VARCHAR(50) UNIQUE NOT NULL,
    email VARCHAR(255) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    badge_number VARCHAR(50) UNIQUE NOT NULL,
    full_name VARCHAR(150) NOT NULL,
    role VARCHAR(30) NOT NULL CHECK (role IN ('INSPECTOR', 'CONTROLLER',
'DIRECTOR', 'ADMIN', 'VIEWER')),
    jurisdiction_id VARCHAR(36) REFERENCES jurisdictions(id) ON DELETE SET NULL,
    is_active BOOLEAN DEFAULT TRUE NOT NULL,
    last_login_at TIMESTAMP WITH TIME ZONE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL,
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);

CREATE INDEX idx_users_role ON users (role);
CREATE INDEX idx_users_badge ON users (badge_number);

-- =====
-- 3. INSPECTIONS TABLE
-- =====

CREATE TABLE inspections (
    id VARCHAR(36) PRIMARY KEY,
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inspection_number VARCHAR(60) UNIQUE NOT NULL, -- Format: INSP-YYYYMMDD-XXXX
officer_id VARCHAR(36) NOT NULL REFERENCES users(id) ON DELETE RESTRICT,
jurisdiction_id VARCHAR(36) NOT NULL REFERENCES jurisdictions(id) ON DELETE
RESTRICT,
capture_source VARCHAR(30) NOT NULL CHECK (capture_source IN ('PHYSICAL_FIELD',
'ECOMMERCE_URL', 'ECOMMERCE_DOM', 'MANUAL_UPLOAD')),
product_name VARCHAR(255) NOT NULL,
brand_name VARCHAR(150),
manufacturer_name VARCHAR(255),
category VARCHAR(100) NOT NULL, -- e.g. 'FOOD_SNACKS', 'COSMETICS', 'COMMODITY'
package_type VARCHAR(50) NOT NULL, -- 'RECTANGULAR', 'CYLINDRICAL',
'FLEXIBLE_POUCH'
ecommerce_url TEXT,
overall_status VARCHAR(20) NOT NULL CHECK (overall_status IN ('PASS', 'FAIL',
'WARNING', 'PENDING_REVIEW')),
ai_verdict VARCHAR(20) NOT NULL CHECK (ai_verdict IN ('PASS', 'FAIL',
'WARNING')),
adjudication_override BOOLEAN DEFAULT FALSE NOT NULL,
adjudication_officer_id VARCHAR(36) REFERENCES users(id) ON DELETE SET NULL,
adjudication_remarks TEXT,
adjudication_timestamp TIMESTAMP WITH TIME ZONE,
gps_latitude DECIMAL(10, 8),
gps_longitude DECIMAL(11, 8),
gps_altitude DECIMAL(8, 2),
device_fingerprint VARCHAR(255) NOT NULL,
inspection_timestamp TIMESTAMP WITH TIME ZONE NOT NULL,
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL,
updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);

```

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CREATE INDEX idx_inspections_number ON inspections (inspection_number);
CREATE INDEX idx_inspections_officer ON inspections (officer_id);
CREATE INDEX idx_inspections_status ON inspections (overall_status);
CREATE INDEX idx_inspections_timestamp ON inspections (inspection_timestamp);

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-- =====
-- 4. EVIDENCE IMAGES TABLE
-- =====

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CREATE TABLE evidence_images (
    id VARCHAR(36) PRIMARY KEY,
    inspection_id VARCHAR(36) NOT NULL REFERENCES inspections(id) ON DELETE
CASCADE,
    panel_type VARCHAR(50) NOT NULL CHECK (panel_type IN ('PDP_FRONT',
'BACK_LABEL', 'SIDE_PANEL', 'TOP_LID', 'BOTTOM', 'ECOMM_GALLERY')),
    file_path TEXT NOT NULL,
    raw_sha256 VARCHAR(64) NOT NULL, -- Immutable forensic SHA-256 hash of raw
input
    image_width INTEGER NOT NULL,
    image_height INTEGER NOT NULL,
    color_channels INTEGER DEFAULT 3 NOT NULL,
    calibration_method VARCHAR(50) NOT NULL CHECK (calibration_method IN
('ARUCO_MARKER', 'STANDARD_COIN', 'FIXED_RIG', 'MANUAL_BENCHMARK', 'UNRESOLVED')),
    calibration_reference_id VARCHAR(50),
    px_to_mm_scale DECIMAL(10, 4), -- e.g. 12.4500 pixels per millimeter
    calibration_error_margin_pct DECIMAL(5, 2), -- e.g. 1.25%
    blur_laplacian_variance DECIMAL(10, 2) NOT NULL,
    glare_pixel_percentage DECIMAL(5, 2) NOT NULL,
    perspective_skew_angle_deg DECIMAL(5, 2),
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);

```

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);
```

```
CREATE INDEX idx_evidence_images_inspection ON evidence_images (inspection_id);
CREATE INDEX idx_evidence_images_hash ON evidence_images (raw_sha256);
```

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-- =====
-- 5. BOUNDING BOXES & EXTRACTED FIELDS TABLE
-- =====
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CREATE TABLE bounding_boxes (
  id VARCHAR(36) PRIMARY KEY,
  image_id VARCHAR(36) NOT NULL REFERENCES evidence_images(id) ON DELETE CASCADE,
  field_type VARCHAR(50) NOT NULL CHECK (field_type IN (
    'MANUFACTURER_ADDRESS', 'PACKER_ADDRESS', 'IMPORTER_ADDRESS',
    'COUNTRY_OF_ORIGIN', 'NET_QUANTITY', 'MRP', 'UNIT_SALE_PRICE',
    'DATE_OF_MANUFACTURE', 'DATE_OF_EXPIRY', 'BEST_BEFORE',
    'CONSUMER_CARE_CONTACT', 'GENERIC_NAME', 'BARCODE_QR'
  )),
  ymin_px INTEGER NOT NULL,
  xmin_px INTEGER NOT NULL,
  ymax_px INTEGER NOT NULL,
  xmax_px INTEGER NOT NULL,
  detection_confidence DECIMAL(5, 4) NOT NULL, -- e.g. 0.9842
  raw_ocr_text TEXT NOT NULL,
  normalized_text TEXT,
  ocr_confidence DECIMAL(5, 4) NOT NULL,
  measured_font_height_px DECIMAL(8, 2),
  measured_font_height_mm DECIMAL(6, 2), -- Derived via px_to_mm_scale
  font_measurement_method VARCHAR(50) DEFAULT 'CONNECTED_COMPONENTS',
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);
```

```
CREATE INDEX idx_bounding_boxes_image ON bounding_boxes (image_id);
CREATE INDEX idx_bounding_boxes_field ON bounding_boxes (field_type);
```

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-- =====
-- 6. COMPLIANCE EVALUATIONS TABLE (RULE ENGINE LEDGER)
-- =====
```

```
CREATE TABLE compliance_evaluations (
  id VARCHAR(36) PRIMARY KEY,
  inspection_id VARCHAR(36) NOT NULL REFERENCES inspections(id) ON DELETE
CASCADE,
  bounding_box_id VARCHAR(36) REFERENCES bounding_boxes(id) ON DELETE SET NULL,
  rule_code VARCHAR(50) NOT NULL, -- e.g. 'RULE_06_1_H_NET_QTY_FONT',
  'RULE_06_1_K_USP'
  rule_legal_citation VARCHAR(255) NOT NULL, -- 'Rule 6(1)(h) read with Table-I,
G.S.R. 629(E)'
  status VARCHAR(20) NOT NULL CHECK (status IN ('PASS', 'FAIL', 'WARNING',
'NOT_APPLICABLE')),
  severity VARCHAR(20) NOT NULL CHECK (severity IN ('CRITICAL', 'MAJOR',
'MINOR')),
  required_value TEXT NOT NULL,
  measured_value TEXT NOT NULL,
  discrepancy TEXT,
  penalty_provision VARCHAR(255) NOT NULL, -- 'Section 36(1) of Legal Metrology
Act, 2009'
  eval_timestamp TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);
```

```
CREATE INDEX idx_compliance_evals_inspection ON compliance_evaluations
```

```

(inspection_id);
CREATE INDEX idx_compliance_evals_rule ON compliance_evaluations (rule_code);
CREATE INDEX idx_compliance_evals_status ON compliance_evaluations (status);

-- =====
-- 7. SECTION 63 BSA 2023 ELECTRONIC CERTIFICATES TABLE
-- =====
CREATE TABLE bsa_certificates (
    id VARCHAR(36) PRIMARY KEY,
    certificate_number VARCHAR(80) UNIQUE NOT NULL, -- Format: CERT-BSA2023-
    YYYMMDD-XXXX
    inspection_id VARCHAR(36) UNIQUE NOT NULL REFERENCES inspections(id) ON DELETE
    RESTRICT,
    issuing_officer_id VARCHAR(36) NOT NULL REFERENCES users(id) ON DELETE
    RESTRICT,
    statutory_law_ref VARCHAR(100) DEFAULT 'Section 63 of Bharatiya Sakshya
    Adhinyam, 2023' NOT NULL,
    device_make_model VARCHAR(150) NOT NULL,
    device_serial_mac VARCHAR(150) NOT NULL,
    operating_system VARCHAR(100) NOT NULL,
    hash_algorithm VARCHAR(30) DEFAULT 'SHA-256' NOT NULL,
    raw_images_merkle_root VARCHAR(64) NOT NULL,
    evidence_bundle_sha256 VARCHAR(64) NOT NULL,
    officer_digital_signature TEXT NOT NULL, -- Base64 encoded RSA/ECDSA signature
    or DSC token
    certificate_pdf_path TEXT NOT NULL,
    generated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);

CREATE INDEX idx_bsa_cert_inspection ON bsa_certificates (inspection_id);
CREATE INDEX idx_bsa_cert_number ON bsa_certificates (certificate_number);

-- =====
-- 8. LEGAL NOTICES TABLE (SECTION 36(1) FORM-1/2 DISPATCH)
-- =====
CREATE TABLE legal_notices (
    id VARCHAR(36) PRIMARY KEY,
    notice_reference_number VARCHAR(100) UNIQUE NOT NULL, -- e.g.
    'LMO/DL/SOUTH/2026/0842'
    inspection_id VARCHAR(36) NOT NULL REFERENCES inspections(id) ON DELETE
    RESTRICT,
    bsa_certificate_id VARCHAR(36) NOT NULL REFERENCES bsa_certificates(id) ON
    DELETE RESTRICT,
    issuing_officer_id VARCHAR(36) NOT NULL REFERENCES users(id) ON DELETE
    RESTRICT,
    recipient_type VARCHAR(50) NOT NULL CHECK (recipient_type IN ('MANUFACTURER',
    'PACKER', 'IMPORTER', 'ECOMMERCE_PLATFORM', 'SELLER')),
    recipient_name VARCHAR(255) NOT NULL,
    recipient_registered_address TEXT NOT NULL,
    recipient_email VARCHAR(255),
    statutory_section VARCHAR(100) DEFAULT 'Section 36(1) of Legal Metrology Act,
    2009' NOT NULL,
    violations_summary TEXT NOT NULL,
    compounding_fee_amount DECIMAL(10, 2), -- e.g. Up to Rs. 25,000 for first
    offence
    reply_window_days INTEGER DEFAULT 15 NOT NULL,
    notice_dispatch_status VARCHAR(30) DEFAULT 'DRAFT' NOT NULL CHECK
    (notice_dispatch_status IN ('DRAFT', 'SIGNED', 'DISPATCHED', 'ACKNOWLEDGED',
    'HEARING_SCHEDULED', 'COMPOUNDED', 'PROSECUTED')),

```

```

generated_pdf_path TEXT NOT NULL,
dispatched_at TIMESTAMP WITH TIME ZONE,
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL,
updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);

CREATE INDEX idx_legal_notices_ref ON legal_notices (notice_reference_number);
CREATE INDEX idx_legal_notices_inspection ON legal_notices (inspection_id);
CREATE INDEX idx_legal_notices_status ON legal_notices (notice_dispatch_status);

-- =====
-- 9. IMMUTABLE AUDIT LOG (CRYPTOGRAPHIC CHAIN / MERKLE CHRONOLOGY)
-- =====

CREATE TABLE audit_logs (
  id VARCHAR(36) PRIMARY KEY,
  sequence_number BIGINT UNIQUE NOT NULL, -- Monotonically increasing sequence
  actor_id VARCHAR(36) NOT NULL REFERENCES users(id) ON DELETE RESTRICT,
  action_type VARCHAR(50) NOT NULL CHECK (action_type IN (
    'USER_LOGIN', 'INSPECTION_CREATED', 'IMAGE_UPLOADED',
    'AI_INFERENCE_EXECUTED', 'OFFICER_OVERRIDE_APPLIED',
    'BSA_CERTIFICATE_ISSUED', 'LEGAL_NOTICE_GENERATED',
    'NOTICE_STATUS_UPDATED', 'DATABASE_EXPORTED'
  )),
  entity_type VARCHAR(50) NOT NULL, -- e.g. 'INSPECTION', 'LEGAL_NOTICE'
  entity_id VARCHAR(36) NOT NULL,
  payload_json TEXT NOT NULL, -- Serialized JSON state snapshot
  previous_hash VARCHAR(64) NOT NULL, -- Hash of previous block
  entry_hash VARCHAR(64) NOT NULL, -- SHA-256(sequence_number || actor_id ||
action_type || payload_json || previous_hash)
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);

CREATE INDEX idx_audit_logs_seq ON audit_logs (sequence_number);
CREATE INDEX idx_audit_logs_actor ON audit_logs (actor_id);
CREATE INDEX idx_audit_logs_entity ON audit_logs (entity_type, entity_id);

```

4. Indexing & Query Optimization Strategy

1. Compound Multi-Tenant & Chronological Indexes:

- Fast filtering for dashboard: (`jurisdiction_id`, `inspection_timestamp DESC`).
- Rapid lookups on legal notices: (`notice_dispatch_status`, `reply_window_days`).

2. Forensic Integrity Constraints:

- `ON DELETE RESTRICT` on all users, inspections, and evidence records associated with issued legal notices or BSA certificates. This guarantees no officer can delete an inspection record after an adverse finding has been signed.

- `raw_sha256` has a unique check constraint per image to prevent redundant identical processing and establish chain of custody.

3. Audit Log Cryptographic Chaining:

- Every row in `audit_logs` includes `previous_hash` and `entry_hash = SHA256(...)`. Any tampering or out-of-band deletion inside SQLite/PostgreSQL causes the Merkle chain verification algorithm to immediately fail.
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5. Data Retention & Archival Policies

- **Inspection Records & Evidence Bundles:**

- Standard retention: Minimum **7 years** (aligned with statutory limitation periods under the Legal Metrology Act and Indian criminal jurisprudence).
- Storage footprint: ~2.5 MB per inspection (1x Raw Full Resolution Image @ 1.8 MB + 4x Cropped Panels @ 400 KB + 1x PDF Evidence Report @ 300 KB).
- 10,000 inspections = ~25 GB storage (easily accommodated on standard NVMe EBS or local storage).